Cross-Domain Meta-Learning Under Dual-Adjustment Mode for Few-Shot Hyperspectral Image Classification

Lei Hu, Wei He, *Senior Member, IEEE*, Liangpei Zhang, *Fellow, IEEE*, and Hongyan Zhang, *Senior Member, IEEE*

***Abstract*— Hyperspectral image (HSI) classification with limited training samples has been well studied in recent years. Among them, the few-shot learning (FSL) technique demonstrates excellent processing capability under limited labeled samples. Nevertheless, the current FSL-based works provide scarce attention to effective class prototypes and metric types, resulting in high generalization error and poor interpretation during the cross-domain testing phase. A dual-adjustment mode-based cross-domain meta-learning (DMCM) method for few-shot HSI classification is proposed to tackle this issue. Specifically, a three-dimensional ghost attention network (TGAN) with strong learning capability without massive parameters is first constructed. Meanwhile, a dual-adjustment mode comprising intracorrection (IC) and interalignment (IA) learning strategies is then adopted to solve domain shift issue via episode-level meta-tasks, where IC and IA focus on effective class prototypes and data distribution differences between domains, respectively. Afterward, considering that the traditional Euclidean distance metric is insensitive to the distribution of within-class samples, the class-covariance metric (CCM) is employed to account for the distribution in feature space of each class to optimize decision boundary and alleviate the misclassification problem. Extensive experiments on three publicly available target hyperspectral datasets demonstrate the effectiveness of the proposed method in comparison with other state-of-the-art (SOTA) methods. The codes will be available on the website at https://github.com/HlEvag/DMCM.git.**

***Index Terms*— Cross domain, few-shot learning (FSL), hyperspectral image (HSI) classification, meta-learning.**

I. Introduction

A HYPERSPECTRAL image (HSI) has been widely applied in environmental monitoring, accurate agriculture, land use, and urban planning due to the abundant spectral and spatial information [1], [2], [3], [4], [5]. In particular, in HSI analysis and applications, the HSI classification has received extensive attention in the last decades [6], [7].

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The early HSI classification methods can be categorized into two parts: feature extraction and classification. Commonly used feature extraction methods include principal component analysis [8], linear discriminant analysis [9], manifold learning [10], and morphological methods [11]. These methods extract critical information from the original HSIs. In addition, the classifiers, such as k-nearest neighbor [12], extreme learning machine [13], and support vector machine (SVM) [14], are adopted to classify each pixel of HSI. Nevertheless, these methods mainly rely on handcrafted features and tend to have a relatively shallow structure, resulting in difficulties in extracting discriminatory features [15].

As an effective deep learning (DL) tool, the convolutional neural network (CNN) has been widely used for the HSI classification task and has shown its excellence [16], [17], [18], [19], [20]. Hu et al. [21] initially introduced a 1-D convolutional kernel to process spectral information in HSIs, which provided a new approach to process HSIs. Spatial information has proved to be extremely effective in HSI classification [22], [23]. Makantasis et al. [24] mined spatial information in HSIs by constructing a 2-D CNN (2DCNN). Chen et al. [25] used a combination of stacked autoencoder (SAE) and deep belief network (DBN), which incorporates spatial and spectral features, to improve the classification accuracy. However, the strength of the unique 3-D cube structure of HSIs was not demonstrated. Li et al. [26] proposed a new 3DCNN framework to capture spatial–spectral features simultaneously efficiently. In addition to these single CNNs, a hybrid network (HybridSN) based on 2-D and 3-D convolutional kernels was designed in [27], which was validated experimentally for its potential in HSIs. Numerous researchers considered ResNet [28] and DenseNet [29] to address the potential problem of gradient disappearance in deep networks. Zhong et al. [30] introduced residual connection to build an end-to-end spatial–spectral residual network (SSRN) for improving the classification performance of the model. Feng et al. [31] further refined HybridSN by adding a residual module (R-HybridSN). Inspired by DenseNet, Wang et al. [32] combined dense connectivity and multiscale convolutional kernels to extract extensive multiscale spatial–spectral features and also spent reduced training time. However, the above methods rely heavily on extensive training of labeled samples, while obtaining sufficient supervised information is time-consuming

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and laborious. Thus, improving the HSI classification accuracy of these DL-based methods with limited training samples still remains as a considerable challenge.

Typically, two general strategies are used to deal with HSI classification with small training samples. The first strategy is to extract additional prior knowledge and features from HSIs through different learning strategies (i.e., supervised learning [33], semisupervised learning [34], and unsupervised learning [35]). Xue et al. [36] suggested a novel spectral–spatial Siamese network for HSI classification, which fed small patches into each branch to collect additional features. However, the classification performance is unstable with fewer labeled sample sizes. Wu and Prasad [37] committed to semisupervised learning by assigning pseudo-labels to unlabeled data to further reduce the demand on labeled samples. Mei et al. [38] implemented HSI classification based on a 3-D convolutional autoencoder using an unsupervised spatial–spectral feature learning strategy. Furthermore, Xu et al. [39] learned meaningful spectral–spatial feature representations from high-level semantic perspective. However, these methods still require real label information to fine-tune the classifier. In addition, attention mechanism [40] and lightweight network structures [41] have gained considerable attention in the feature extraction phase. Roy et al. [42] combined the squeeze and excitation network (SENet) with residual units to capture the relationship between channels. Inspired by the convolutional block attention module (CBAM), the spatial–spectral attention module was embedded into the residual block to avoid overfitting [43]. These attention modules are effective in highlighting key features, but their structures are redundant. Impressively, compared with SENet and CBAM, normalization-based attention (NAM) [44] generated the same emphatic level without additional cost. A lightweight network-based depthwise separable convolution was designed for deep discriminative feature learning and classification [45]. Inspired by GhostNet, Paoletti et al. [46] received positive feedback on using ghost modules, which generate additional features by using a few parameters. The aforementioned DL-based methods achieved some progress, but are still limited in their classification accuracy.

The second strategy is introducing the concept of transfer learning [47]. First, traditional transfer learning methods attempt to transfer knowledge learned from one domain to another. Two data domains are generally available: source domain with abundant labeled samples and target domain containing few labeled samples. Yang et al. [48] are the first to employ transfer learning in the HSI classification task. First, a two-branch spatial–spectral feature extractor is trained on the source dataset and then fine-tuned by small, labeled samples of the target dataset. The later transfer learning models based on fine-tuning basically followed this architecture [49], [50]. However, using these transfer learning methods to address the new classes in the transfer stage is difficult when the difference in data distribution between two domains is substantial. In practical HSI classification applications, the target classes are not only poorly represented by the few training samples, but may also come from different other domains [51]. For example, the source and target datasets could be captured by different acquisition conditions (e.g., various sensor type, and weather and temperature change), leading to different data distributions. Fortunately, this problem can be tackled by few-shot learning (FSL) [52]. FSL is essentially a meta-learning method, which is effectively adapted to hidden classes given a few labeled samples [53], [54], [55], [56]. Notably, the labeled target samples are usually selected around five per class. Recent researchers have mainly favored the few-shot HSI classification approach based on metric learning, where classification is accomplished by measuring the distance between the query and the support sets [57], [58], [59]. Liu et al. [60] trained a deep residual 3DCNN to learn a metric space by episodic learning mode, and experiments proved the potential of FSL for the HSI classification with rarely labeled samples. The relation network for HSI few-shot classification (RN-FSC) [61] method was presented to approximate the relationship between sample features based on deep relational network. However, these methods neglect the domain shift problem due to the different data distributions in the source and target domains, which affects the generalization capability of the model [62]. Li et al. [63] first considered cross-domain issues in the few-shot HSI classification field. Furthermore, Zhang et al. [64] introduced domain alignment based on graph information aggregation to explore nonlocal spatial information with strong correspondence.

The solution in this article is established on the concept of FSL. Current FSL-based methods have achieved remarkable progress in HSI classification with limited training samples but still encountered the following drawbacks.

1) Overfitting is a common issue in few-shot HSI classification, which is exacerbated by the scarcity of labeled data. This dilemma is primarily due to the overfitting of the feature extractor with a large number of trainable parameters to the training examples, which can lead to poor generalization to new examples. Therefore, designing a low-parameter feature extractor, which can learn powerful representations with limited data, is a significant challenge in few-shot HSI classification.
2) Sensor types and acquisition conditions may differ between the source and target domains. Thus, it is easy to get stuck in the domain shift dilemma when performing cross-domain FSL (DCFSL) tasks. At present, FSL-based methods mainly concentrate on using support sample prototypes directly to achieve accurate prediction of query samples, while disregarding the effectiveness of class prototypes. Consequently, this phenomenon may lead to subsequent prototype-based inference errors.
3) The mainstream FSL methods are based on metric learning and generally adopt Euclidean distance as the metric strategy. Nevertheless, the traditional Euclidean distance metric is insensitive to the distribution of within-class samples, which can easily cause serious misclassification issues. A metric approach that can improve the nonlinear classifier decision boundary is urgently needed.

In this article, a dual-adjustment mode-based cross-domain meta-learning (DMCM) method is proposed for few-shot HSI

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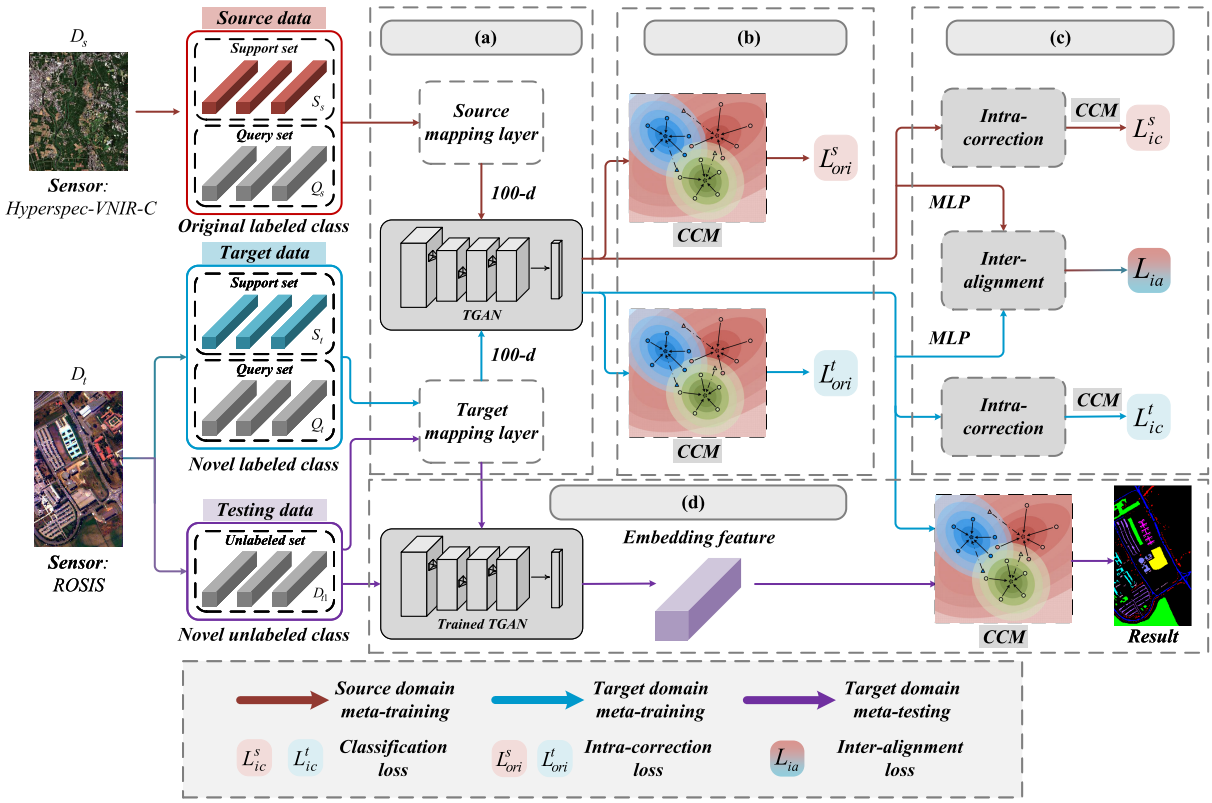


Fig. 1. Overall framework of the DMCM. The DMCM framework comprises the following four parts. (a) Feature extraction. (b) DCFSL. (c) Dual-adjustment mode. (d) Target dataset classification.

classification. The main idea is to obtain discriminant feature representation in the meta-training phase through a robust extractor and a dual-adjustment learning strategy. Comprehensively, a three-dimensional ghost attention network (TGAN), which is combined with ghost blocks and NAM, is first established to obtain additional discriminative embedding features using small number of training parameters. A dual-adjustment mode comprising intracorrection (IC) and interalignment (IA) strategies is then applied to enhance model generalization, where IC and IA are used to regulate the generation of valid class prototypes and resolve the domain shift problem, respectively. Finally, the class-covariance metric (CCM) is used to improve nonlinear classifier decision boundaries and reduce misclassification issues by considering the feature space distribution of each class. In general, extensive experiments have proven that the proposed method can achieve highly competitive performance on few-shot HSI classification task. Notably, the contributions can be summarized as follows.

1) A low-parametric feature extractor TGAN based on the GA modules, which enables obtaining additional discriminative features by adaptively updating few parameters in the meta-tasks learning phase, is introduced.
2) A dual-adjustment mode is proposed for few-shot HSI classification task. Compared with traditional FSL methods, the proposed method not only applies the IA strategy based on maximum mean discrepancy (MMD) to alleviate the domain shift problem but also regulates the generation of highly efficient class prototypes by the IC approach.
3) The CCM is introduced to acquire the distribution of the feature space of each class, improving the decision boundary of the nonlinear classifier and enhancing the classification capability of the model.

The rest of this article is organized as follows. Section II elaborates on the proposed DMCM with details. Section III exhibits the extensive experiments, and Section IV discusses the proposed method. Section V contains a summary of this article and an outlook on the direction of future efforts.

II. Methodology

A. Overall Framework

The overall framework of the proposed DMCM is illustrated in Fig. 1. Specifically, DMCM comprises the following four main parts: feature extraction, DCFSL, dual-adjustment mode, and target dataset classification. The first stage is feature extraction, namely, a TGAN feature extractor, which combines the ghost block and NAM, is designed to obtain the spatial–spectral embedding representation. Source-domain FSL and target-domain FSL are then performed alternatively to mine the source class efficiently for transferable knowledge and learn the discriminative embedding representation of the target class. Notably, the CCM is adopted to measure the potential relationship between support and query samples. In addition, the dual-adjustment mode comprises IC and IA, in which IC can generate useful class prototypes, and the interdomain data distribution differences can be reduced by IA. In the final target dataset classification stage, the trained

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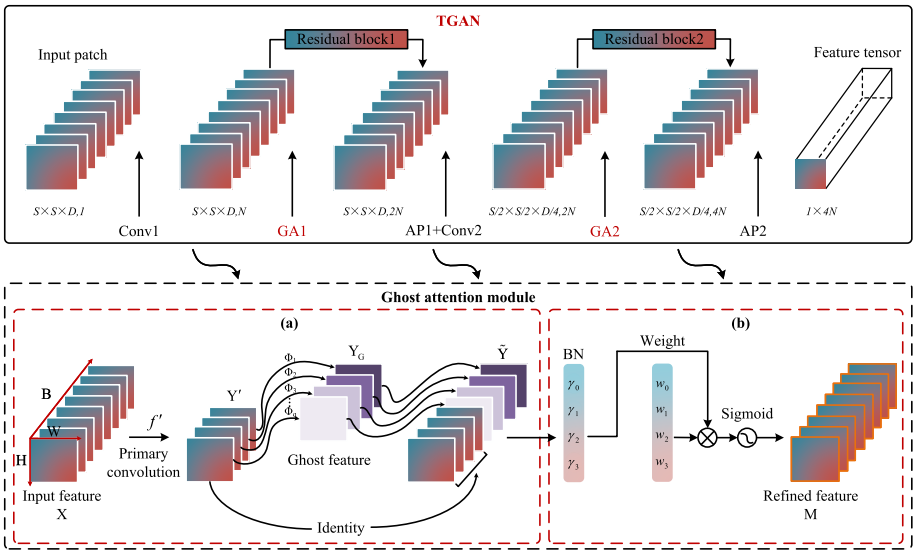


Fig. 2. Structure of the TGAN proposed in this article, which comprises ghost block, NAM block, and two residual blocks. The goal of TGAN is to obtain additional discriminable feature representations without massive parameters. (a) Ghost block. (b) NAM block.

TGAN, which can extract domain-invariant features, is applied to classify unlabeled test dataset. The details of the DCFSL training strategy will be presented subsequently.

B. Feature Extractor Based on GA Block

1) Ghost Module for Additional Embedding Features: Inspired by the literature [65], the idea of "ghost" is introduced into the traditional 3-D convolution operation to generate additional feature maps using cheap operations to reduce the number of trainable parameters, and improve the efficiency of the model. The schematic of the ghost module is shown in Fig. 2(a).

In practice, given the original HSI data $\mathbf{X} \in \mathbb{R}^{H\times W\times B}$, H, W, and B denote the height, width, and number of spectral bands, respectively. The procedure of processing X using a classical 3-D convolutional layer can be simply formulated as follows:

$$\mathbf{Y} = \mathbf{X} * \mathbf{f} + \mathbf{b} \tag{1}$$

where $\mathbf{Y} \in \mathbb{R}^{H'\times W'\times B\times n}$ represents the output feature map with n channels in the 3-D convolutional layer, and H' and W' denote the height and width of the output data, respectively. $\mathbf{f} \in \mathbb{R}^{k\times k\times B\times n}$ denotes the 3-D convolutional filter with a kernel size of $k \times k$, and b is the bias.

In this article, the original 3-D convolutional layer is decomposed into two phases of convolutional operations. Specifically, the first stage involves ordinary convolution, but the total number of channels will be strictly controlled. Ordinary convolution filters $\mathbf{f}' \in \mathbb{R}^{k\times k\times B\times n'}$ are applied to generate the middle output feature maps $\mathbf{Y}' \in \mathbb{R}^{H'\times W'\times B\times n'}$, where $n' \le n$. In the second stage, the cheap operation is performed on each intrinsic feature in $\mathbf{Y}'$ to acquire g ghost features to obtain the desired n feature maps as in (1). The above operation can be represented by the mathematical formula as follows:

$$\mathbf{Y}' = \mathbf{X} * \mathbf{f}' + \mathbf{b}' \tag{2}$$

$$\mathbf{Y_G} = \left[\mathbf{\Phi}_{i,j}\left(\mathbf{y}'_i\right)\right] \quad \forall\, i = 1,\ldots,n' \quad j = 1,\ldots,g \tag{3}$$

$$\tilde{\mathbf{Y}} = \mathrm{Cat}\left[\mathbf{Y}', \mathbf{Y_G}\right] \tag{4}$$

where $\tilde{\mathbf{Y}}$ represents the output feature map with $n = n' \cdot g$ channels, $\mathbf{y}'_i$ is the ith intrinsic feature map in $\tilde{\mathbf{Y}}$, and $\mathbf{\Phi}_{i,j}$ indicates the jth linear operation to obtain the jth ghost feature map from the ith intrinsic feature map. Surprisingly, $\mathbf{y}'_i$ can produce one or more ghost feature maps. In addition, compared with the original convolution, the linear operation $\mathbf{\Phi}$ is less computationally intensive.

2) NAM Module: NAM is used to reassign feature weights to highlight key features and suppress less significant channels to extract discriminative features further. Remarkably, compared with the popular SE and CBAM, NAM does not involve redundant convolution and full connection operations, remaining highly computationally efficient while maintaining similar performance [44]. Hence, a batch normalization (BN) scale factor is employed in the channel direction, as shown in (5), where the scale factor measures the variance of the channel and indicates its significance

$$B_{\text{out}} = \mathrm{BN}(B_{\text{in}}) = \gamma \frac{B_{\text{in}} - \mu_\rho}{\sigma_\rho^2} + \beta \tag{5}$$

where μ_ρ and σ_ρ are the mean and standard deviation of small batch ρ, respectively, and γ and β denote the scaling and shift factors, respectively, which are trainable transformation parameters. The detailed structure of the NAM is illustrated in Fig. 2(b), and the output feature M obtained can be expressed as follows:

$$M = \mathrm{Sigmoid}\left(W_\gamma\left(\mathrm{BN}\left(\tilde{\mathbf{Y}}\right)\right)\right) \tag{6}$$

where γ is the scaling factor in the channel direction and $W_\gamma = \gamma_i / \sum_{j=0} \gamma_j$ denotes the obtained weights.

3) Proposed TGAN Architecture for Reducing Parameters and Operational Redundancy: The TGAN framework proposed in this article is presented in Fig. 2. The TGAN structure generally mainly comprises normal convolutional

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TABLE I
CONFIGURATIONS OF THE EMBEDDING FEATURE EXTRACTOR TGAN

Layer name		Kernel shape	Output shape	Connected to
Input layer		-	(9 × 9 × 100, 1)	-
Conv1:	Conv3D-BN-ReLU	(3, 3, 3)	(9 × 9 × 100, N)	Input layer
GA1:	Conv3D-BN-ReLU	(1, 1, 1)	(9 × 9 × 100, 2N/r)	Conv1
	Conv3D-BN-ReLU	(3, 3, 3)	(9×9×100, 2N × (r-1)/r)	
	Concatenate	-	(9 × 9 × 100, 2N)	
	NAM	-	(9 × 9 × 100, 2N)	
Residual block1:	Conv3D-BN	(1, 1, 1)	(9 × 9 × 100, 2N)	Conv1
	Add	-	-	
AP1+Conv2:	AvgPool3D	(4, 2, 2)	(4 × 4 × 25, 2N)	Residual block1
	Conv3D-BN-ReLU	(1, 1, 1)	(4 × 4 × 25, 2N)	
GA2:	Conv3D-BN-ReLU	(1, 1, 1)	(4 × 4 × 25, 4N/r)	AP1+Conv2
	Conv3D-BN-ReLU	(3, 3, 3)	(4 × 4 × 25, 4N × (r-1)/r)	
	Concatenate	-	(4 × 4 × 25, 4N)	
	NAM	-	(4 × 4 × 25, 4N)	
Residual block2:	Conv3D-BN	(1, 1, 1)	(4 × 4 × 25, 4N)	AP1+Conv2
	Add	-	-	
AP2:	AvgPool3D	(4, 2, 2)	(2 × 2 × 6, 4N)	Residual block2
Flatten:	-	-	1 × 4N	AP2

layers (Conv1 and Conv2), average pooling layers (AP1 and AP2), two residual blocks, and two GA modules, in which the GA module is formed by connecting the ghost block to the NAM head and tail, obtaining rich key features under few training parameters. The configuration details of TGAN are shown in Table I. Specifically, the publicly available hyperspectral dataset is assumed as $\mathbf{X} \in \mathbb{R}^{H\times W\times B}$ to adapt the training process of the network, where X is decomposed into 3-D cube structure with $S \times S$ window size. The embedding features of each cube are eventually obtained using TGAN. However, the number of bands in the source- and target-domain datasets may be different. Thus, the band numbers are uniformly mapped to D dimensions by a mapping layer before TGAN, and S and D are, respectively, set to 9 and 100 in this article [63], [64]. Significantly, the performance of TGAN is closely related to the number of 3-D convolution kernels N and the scaling factor r in the ghost module, which are thoroughly discussed in Section III.

C. DCFSL Training Strategy

The meta-learning training strategy is adopted to execute FSL in the source and target domains, which can discover the transferable knowledge in the source class and learn the distinguished embedding model from the target class [66], [67]. The source dataset (base dataset) and target dataset (novel dataset) are assumed to be D_s (contain C_s classes) and D_t (contain C_t classes), respectively, and $D_s \bigcap D_t = \emptyset$, that is, the source and target domains do not contain the same class. The FSL training process in the source domain can be regarded as several small meta-tasks (i.e., episodes). After obtaining the spatial–spectral embedding features of the samples, the FSL of source domain is performed for each episode by calculating the distance between the embedding features of unlabeled samples and the class prototypes of labeled samples. First, M_s classes are randomly selected from the C_s classes of source domain, and each class contained K label samples as support set $S_s = \{(x_{is}, y_{is})\}_{is=1}^{M_s\times K}$. Then, T unlabeled samples are randomly selected from the remainder of the same M_s classes to form the query set $Q_s = \{(x_{js}, y_{js})\}_{js=1}^{M_s\times T}$. The class distribution P_φ^s of sample x_{js} in Q_s is expressed as follows:

$$P_\varphi^s\left(y_{js}=k\middle|x_{js}\in Q_s\right)=\frac{\exp\left(-d_k\left(f_\varphi\left(x_{js}\right),P_k^s\right)\right)}{\sum_{k=1}^{M_s}\exp\left(-d\left(f_\varphi\left(x_{js}\right),P_k^s\right)\right)} \tag{7}$$

$$P_k^s=\frac{1}{K}\sum_{x_{is}\in S_s} f_\varphi(x_{is}) \tag{8}$$

where P_k^s is the kth class prototype in S_s, f_φ represents the embedded feature extractor with parameters φ, x_{js} and y_{js} denote the kth class samples and labels, respectively, and d_k is the distance metric approach. Therefore, the loss function of the FSL on source domain is defined as follows:

$$L_{\text{ori}}^s=E_{S_s,Q_s}\left[-\sum_{(x,y)\in Q_s}\log P_\varphi^s\left(y_{js}=k\middle|x_{js}\right)\right]. \tag{9}$$

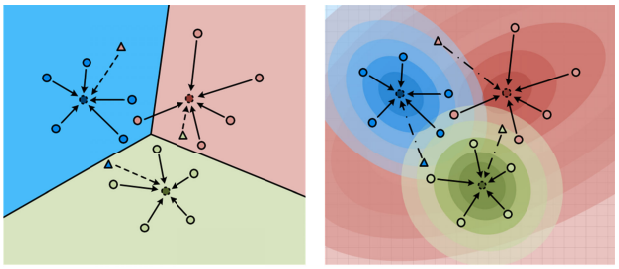
The FSL process for the target domain is similar to source domain, and the main difference lies in the division of D_t into a labeled training set D_L and an unlabeled testing set D_{t1}. Significantly, D_L is a few-shot dataset comprising L samples from each class, which is insufficient to form a single training episode. Therefore, the addition of random noise and flipping strategies is adopted to augment the D_L data. The support and query sets are denoted as $S_t = \{(x_{it}, y_{it})\}_{it=1}^{M_t\times K}$ and $Q_t = \{(x_{jt}, y_{jt})\}_{jt=1}^{M_t\times T}$, respectively, and the class distribution P_φ^t of the sample x_{jt} in Q_t can be formulated as follows:

$$P_\varphi^t\left(y_{jt}=k\middle|x_{jt}\in Q_t\right)=\frac{\exp\left(-d_k\left(f_\varphi\left(x_{jt}\right),P_k^t\right)\right)}{\sum_{k=1}^{M_t}\exp\left(-d\left(f_\varphi\left(x_{jt}\right),P_k^t\right)\right)} \tag{10}$$

$$P_k^t=\frac{1}{K}\sum_{x_{it}\in S_s} f_\varphi(x_{it}) \tag{11}$$

where P_k^t is the class prototype of the kth class in S_t, and x_{jt} and y_{jt} denote the samples and labels of the kth class in Q_t, respectively. The loss function of the target domain can

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(a) (b)

Fig. 3. Schematic of the different metrics. Different color circles indicate the embedding features of different classes in the support set, and triangles are the embedding space representation of the query set to be tested. (a) Euclidean distance metric. (b) CCM.

be expressed as follows:

$$L_{\text{ori}}^{t} = E_{S_t,Q_t}\left[-\sum_{(x,y)\in Q_t}\log P_{\varphi}^{t}\left(y_{jt}=k|x_{jt}\right)\right]. \tag{12}$$

Overall, the loss function of a standard DCFSL can be represented by the following equation:

$$L_{\text{fsl}} = L_{\text{ori}}^{s} + L_{\text{ori}}^{t}. \tag{13}$$

D. Dual-Adjustment Mode

1) Class-Covariance-Based Metric: In the few-shot HSI classification field, substantially less attention has been focused on the metric for computing distances in the embedding space [68]. Although the Euclidean distance metric has been proven to be better than the traditional cosine metric, the performance of this metric is based on the flawed assumptions of the following: 1) uncorrelated feature dimensions and 2) uniform variance. Moreover, this metric is insensitive to the distribution of within-class samples [69]. As shown in Fig. 3(a), the Euclidean distance is bounded by the distribution of features and suffers from misclassification. Therefore, CCM is adopted to consider the distribution of each class in the feature space and improve the decision boundary of the nonlinear classifier to alleviate the misclassification problems, as illustrated in Fig. 3(b). The distance metric formula is presented as follows:

$$d_k(x,y) = \frac{1}{2}(x-y)^T(\mathbf{Q}_k)^{-1}(x-y) \tag{14}$$

where $\mathbf{Q}_k$ denotes the covariance matrix of the kth class of the current task. $\mathbf{Q}_k$ cannot be determined in advance; therefore, $\mathbf{Q}_k$ is estimated in each meta-task using the embedding features of the support set. Concretely, $\mathbf{Q}_k$ is formed by the intra-CCM Σ_k and the inter-CCM Σ, which can be formulated as follows:

$$\mathbf{Q}_k = \lambda_k\Sigma_k + \left(1-\lambda_k\right)\Sigma + I \tag{15}$$

where Σ_k is generated using the embedding features of the kth class dataset S_k in the support set and is implemented as follows:

$$\Sigma_k = \frac{1}{|S_k|-1}\sum_{(x_i,y_i)\in S_k}\left(f_{\varphi}(x_i)-\mu_k\right)\left(f_{\varphi}(x_i)-\mu_k\right)^T \tag{16}$$

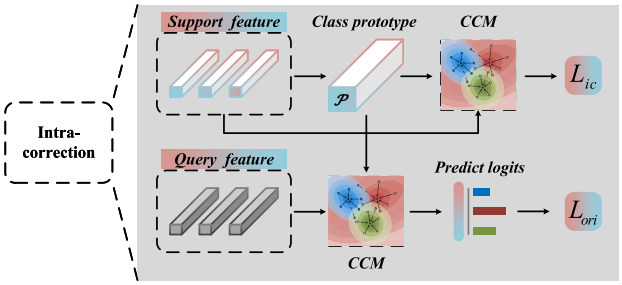


Fig. 4. Flowchart of the IC learning strategy.

where f_{φ} indicates the feature extractor and μ_k is the current task kth class mean vector. If $|S_k| = 1$, then Σ_k will be defined as a suitable zero-size matrix. Similar to (16), Σ is obtained by using the entire support samples while ignoring their categories. $\lambda_k = |S_k|/(|S_k|+1)$ represents the weighted estimate of the covariance matrix Σ_k.

2) IC Learning Strategy for Effective Class Prototypes: Most existing prototype-based learning methods have recently mainly adopted the paradigm of directly predicting query samples using the prototypes of support samples, which focuses on precision performance in the query set and ignores the validity of the prototypes [70]. Instead of the typical FSL technique, an IC learning approach is applied to maximize the information in the support set. This approach produces considerably robust class prototypes by adding predicted loss from support samples to the original loss. Details of the IC learning architecture are illustrated in Fig. 4. The loss of source-domain support samples x_{is} and target-domain support samples x_{it} can be, respectively, expressed as follows:

$$L_{\text{ic}}^{s} = E_s\left[-\sum_{(x,y)\in S_s}\log P_{\varphi}^{s}(y_{is}=k|x_{is})\right] \tag{17}$$

$$L_{\text{ic}}^{t} = E_t\left[-\sum_{(x,y)\in S_t}\log P_{\varphi}^{t}(y_{it}=k|x_{it})\right] \tag{18}$$

where L_{ic}^{s} and L_{ic}^{t} represent the support samples classification loss of source and target domains, respectively. Thus, the original classification losses for different domains can be updated as follows:

$$L_{\text{fsl}}^{s} = L_{\text{ori}}^{s} + L_{\text{ic}}^{s} \tag{19}$$

$$L_{\text{fsl}}^{t} = L_{\text{ori}}^{t} + L_{\text{ic}}^{t}. \tag{20}$$

3) IA Learning Strategy for Domain Adaptation: The model trained on source dataset cannot be directly applied to target domain due to the different acquisition conditions, such as geography, environment, and sensor type, resulting in interdomain data distribution differences [71]. Among the available methods, the MMD is an effective nonparametric distance measure, which is one of the most widely used strategies to measure the distribution differences between the source and target domains [72]. First, the embedded features obtained by TGAN in this article are inputted into MLP to gain class probability representation. MMD is then applied to align the different domains. The IA learning strategy flowchart is shown

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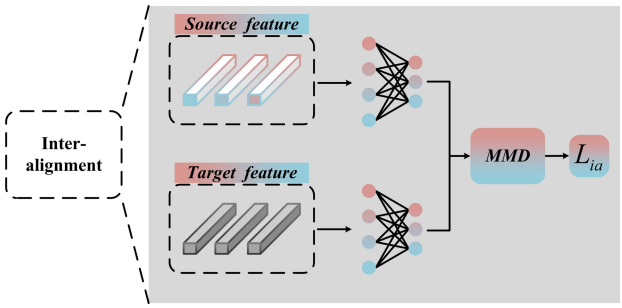


Fig. 5. Flowchart of the IA learning strategy.

in Fig. 5. The loss of MMD is illustrated as follows:

$$L_{\text{ia}} = \left\| \frac{1}{n_s} \sum_{i=1}^{n_s} \phi\left(x_i^s\right) - \frac{1}{n_t} \sum_{j=1}^{n_t} \phi\left(x_j^t\right) \right\|_H^2 \tag{21}$$

where n_s and n_t are the number of samples in source and target domains, respectively, and $\phi()$ is a learning transformation for projecting the original samples into reproducing kernel Hilbert space (RKHS). Significantly, the equation aims to calculate the average difference between the source and target domains of the RKHS, with small MMD values indicating the similarity of the two domains, and vice versa.

Therefore, the total loss function of the source domain can be presented as follows:

$$L^s = L_{\text{fsl}}^s + L_{\text{ia}} = \lambda\left(L_{\text{ori}}^s + L_{\text{ic}}^s\right) + L_{\text{ia}} \tag{22}$$

where λ is a hyperparameter. Similarly, the total loss function of the target domain can be presented as follows:

$$L^t = L_{\text{fsl}}^t + L_{\text{ia}} = \lambda\left(L_{\text{ori}}^t + L_{\text{ic}}^t\right) + L_{\text{ia}}. \tag{23}$$

Notably, FSL is performed on the source and target domains alternately.

III. Experiments

A. Data Description

The experimental datasets used in this article mainly comprise source and target datasets, where the source dataset uses Chikusei scene with rich tagging information (base classes) and the target dataset has limited labeled samples (novel classes), including Indian Pines (IP), University of Pavia (UP), and Pavia Center (PC) scenes. Each dataset is comprehensively described in this section.

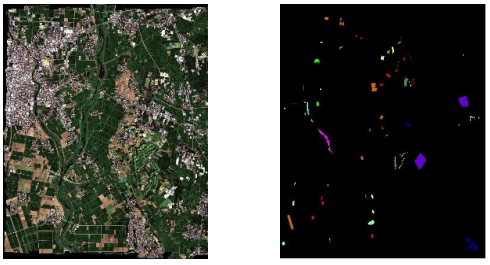
1) Source-Domain Dataset: The Chikusei dataset was collected by the Hyperspectral Visible/Near-Infrared C-Series (Hyperspec-VNIR-C) sensor in Chikusei, Japan, in 2014. This dataset comprises 128 bands, which range from 368 to 1018 nm with 2.5-m resolution. Chikusei has a spatial size of 2517 × 2335 pixels and 19 classes, including water, soil, and vegetation, providing abundant information for cross-domain model training. Tables II and III present detailed attribute information and the number of samples per class for the Chikusei dataset, respectively. Meanwhile, Fig. 6 shows false-color image and ground-truth map of the Chikusei dataset.

TABLE II
Detailed Description of the Source-Domain Dataset Attributes

Properties	Chikusei
Spatial size	2517 × 2335
Wavelength range (nm)	368-1018
Bands	128
Spatial resolution (m/pixel)	2.5
Sensor	Hyperspec-VNIR-C
Classes	19

TABLE III
Number of the Samples for Each Class in the Chikusei Dataset

Label	Color	Class name	Samples
1		Water	2845
2		Bare soil (school)	2859
3		Bare soil (park)	286
4		Bare soil (farmland)	48525
5		Natural plants	4297
6		Weeds in farmland	1108
7		Forest	20516
8		Grass	6515
9		Rice field (grown)	13369
10		Rice field (first stage)	1268
11		Row crops	5961
12		Plastic house	2193
13		Manmade (non-dark)	1220
14		Manmade (dark)	7664
15		Manmade (blue)	431
16		Manmade (red)	222
17		Manmade grass	1040
18		Asphalt	801
19		Paved ground	145
		Total	77,592



(a) (b)

Fig. 6. (a) False-color RGB image and (b) ground-truth map for the Chikusei scene.

2) Target-Domain Dataset: The IP, UP, and PC datasets are selected as target datasets. The study areas for these datasets are located in woodlands, campuses, and cities, containing typical features, such as plants, buildings, and cultivated land. Table IV provides detailed descriptions of the different target datasets.

The IP dataset was obtained by the Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) from IN, USA, in 1992. This dataset has a spatial size of 145 × 145 pixels with 20-m spatial resolution and contains 16 classes and 10 249 samples. The IP dataset has a wavelength range of 400–2500 nm and 220 bands. The experiments are conducted using the remaining spectral bands, excluding the 20 bands affected by noise. The number of samples per class and

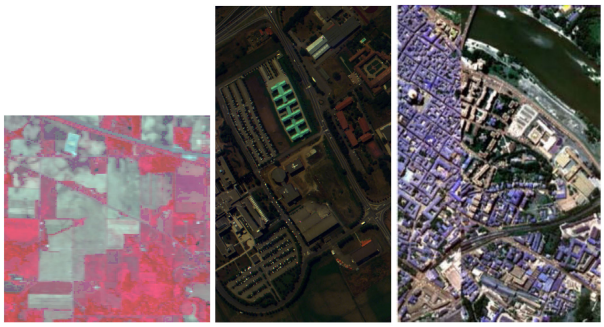
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TABLE IV
DETAILED DESCRIPTION OF THE THREE TARGET-DOMAIN DATASET ATTRIBUTES

Datasets	Spatial size (pixels)	Wavelength range (nm)	Bands	Spatial resolution (m/pixel)	Sensor	Classes
IP	145×145	400-2500	200	20	AVIRIS	16
UP	610×340	430-860	103	1.3	ROSIS	9
PC	1096×715	430-860	102	1.3	ROSIS	9

TABLE V
NUMBER OF THE SAMPLES FOR EACH CLASS IN DIFFERENT TARGET DATASETS

Label	Color	IP		UP		PC	
		Class name	Samples	Class name	Samples	Class name	Samples
1		Alfalfa	46	Alfalfa	6631	Water	65971
2		Corn-notill	1428	Meadows	18649	Trees	7598
3		Corn-mintill	830	Gravel	2099	Asphalt	3090
4		Corn	237	Trees	3064	Self-Blocking Bricks	2685
5		Grass-pasture	483	Painted metal sheets	1345	Bitumen	6584
6		Grass-trees	730	Bare Soil	5029	Tiles	9248
7		Grass-pasture-mowed	28	Bitumen	1330	Shadows	7287
8		Hay-windrowed	478	Self-Blocking Bricks	3682	Meadows	42826
9		Oats	20	Shadows	947	Bare Soil	2862
10		Soybean-notill	972				
11		Soybean-mintill	2455				
12		Soybean-clean	593				
13		Wheat	205				
14		Woods	1265				
15		Buildings-Grass-Trees-Drives	386				
16		Stone-Steel-Towers	93				
		Total	10,249		42,776		148,152



(a) (b) (c)

Fig. 7. False-color RGB image. (a) IP scene. (b) UP scene. (c) PC scene.

visualization of the IP dataset are presented in Table V and Fig. 7(a), respectively.

The UP dataset was acquired by the Reflective Optics Spectrographic Imaging System (ROSIS-03) at the UP campus, Italy, in 2003. The spatial resolution of this dataset is 1.3 m and contains 610 × 340 pixels, nine classes, and 42 776 samples in total. The wavelength range of the UP dataset is 430–860 nm, and 103 spectral bands are applied by removing the 12 bands damaged by noise. Details of the samples and visualization maps of the UP dataset are shown in Table V and Fig. 7(b), respectively.

The PC dataset, which was also captured by the ROSIS sensor with 102 bands, has a spatial size of 1096 × 715 pixels and a wavelength range of 430–860 nm. The sample distribution and visualization maps are, respectively, displayed in Table V and Fig. 7(c).

B. Experimental Setting

1) Implementation Details: A total of 200 labeled samples of each class are randomly selected from the original Chikusei dataset for source dataset generation to learn the transferable knowledge. Meanwhile, $L = 5$ labeled samples per class are randomly selected from the target datasets for training. The number of bands D and window size S of the input hyperspectral patch are set to 100 and 9 uniformly and the number of 3-D convolutional filters N and the scaling factor r in the ghost module to 8 and 4, respectively, to adapt the smooth learning of the feature extractor TGAN. The training iterations were set to 10 000, and the learning rate was set to 0.001 to train the DMCM sufficiently. A cross-entropy loss function can adopted to measure the error, and an Adam optimizer is used to improve the training process. In addition, during the FSL of source domain and target domain, we decompose the entire HSI classification task into many C-way, K-shot classification tasks, where C is the number of categories of support set samples per episode, and each class contains K samples. C is assigned the number of classes for target datasets (16 for the IP, and nine for UP and PC), and K is set to 1 to simulate FSL. For each episode, 19 samples are randomly selected from the remaining samples of the C classes as unlabeled query samples T. Although the higher the value of T, the better the model can simulate the classification task of the test set, and the limitations of computational efficiency and manually labeled samples must be considered [60].

Considering hardware environment, the experiments were conducted on an NVIDIA server with a Titan RTX 3090 GPU. Furthermore, the primary software configuration included Python-3.9.7 compiler and DL framework PyTorch-1.12.0.

2) Comparison Methods and Evaluation Metrics: Eight HSI classification methods, including three traditional nonmeta-learning methods, such as radial basis function (RBF)-SVM [14], 3DCNN [26], and SSRN [30], are selected for comparison in this article to evaluate the effectiveness of

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TABLE VI
CLASSIFICATION RESULTS OF VARIOUS MODELS ON THE IP DATASET

Label	RBF-SVM [14]	3DCNN [26]	SSRN [30]	DFSL [60]	RN-FSC [61]	DCFSL [63]	Gia-CFSL [64]	CMFSL [68]	DMCM
1	71.71	95.12	59.53	96.75	96.34	88.78	89.27	98.78	**99.02**
2	35.84	37.70	**61.69**	36.38	46.13	42.30	49.41	48.09	61.08
3	36.22	19.77	48.37	38.34	40.61	47.13	49.70	57.35	**66.47**
4	38.10	32.51	45.53	77.16	58.62	82.33	77.11	84.35	**96.08**
5	71.72	88.45	**92.95**	73.92	64.96	75.65	74.39	76.07	90.36
6	69.43	73.65	**90.29**	86.25	69.45	80.08	81.77	86.00	90.23
7	91.30	81.82	60.11	97.10	**100.00**	**100.00**	99.13	**100.00**	99.57
8	76.19	53.35	**99.90**	81.82	77.70	87.02	90.30	87.67	84.76
9	85.33	**100.00**	39.58	75.56	**100.00**	**100.00**	98.00	99.33	**100.00**
10	43.04	41.35	42.20	52.22	25.49	**61.70**	60.46	58.51	60.83
11	24.18	66.71	68.84	59.96	65.51	59.98	61.09	59.59	**72.06**
12	29.49	37.40	48.16	36.56	27.13	42.86	47.33	54.18	**60.31**
13	89.60	85.71	98.66	98.00	**99.75**	99.30	97.45	98.55	98.44
14	61.84	62.57	**99.98**	84.63	76.35	88.57	79.81	86.02	84.44
15	30.60	56.42	46.72	74.10	74.10	79.48	72.18	**82.91**	81.21
16	86.14	90.36	75.39	**100.00**	**100.00**	97.73	99.43	99.20	**100.00**
OA	44.25±4.91	54.76±0.03	64.02±1.23	61.69±1.85	58.17±0.02	65.21±2.38	65.53±2.90	67.48±2.99	**75.70±1.42**
AA	58.80±3.74	63.93±0.02	59.82±1.37	73.05±0.84	69.90±0.33	77.06±1.92	76.68±1.86	79.79±1.20	**84.66±1.17**
Kappa	38.19±5.36	48.72±0.03	64.87±0.72	56.78±1.90	52.52±1.53	60.88±2.55	61.11±3.07	63.40±3.11	**72.43±1.56**

the proposed DMCM. Five meta-learning-based HSI classification methods included typical deep FSL (DFSL) [60], deep relation network (RN-FSC) [61], DCFSL [63], graph information aggregation cross-domain few-shot learning (Gia-CFSL) [64], and FSL framework with a class-covariance metric (CMFSL) (Mahalanobis distance) [68]. The RBF-SVM is implemented by the LibSVM toolbox, with the penalty factor c and the parameter *gamma* obtained by the ten-time cross-validation method. The parameters for 3DCNN and SSRN are set in accordance with the corresponding literature defaults. The Chikusei dataset is adopted as the source-domain dataset for the HSI classification methods based on meta-learning to preserve the fairness of the evaluation, and the same random sampling factors are applied for sample selection. The remaining settings use the default values from open-source code and the original paper. Moreover, the overall accuracy (OA), average accuracy (AA), and Kappa coefficient ($100 \times$ Kappa is used in this article) based on the confusion matrix are selected as evaluation metrics in this article.

C. Classification Performance

Three publicly available datasets are employed for classification experiments in this section to obtain the classification performance of various models on different target-domain datasets. The experimental results contain the average of ten results. The highest classification accuracy is displayed in bold, and the second highest accuracy is underlined.

1) Classification Maps and Results of the IP Dataset: Table VI displays the experimental results of the various classification methods on the IP dataset. As shown in Table VI, compared with other methods, the proposed DMCM method has significantly higher OA, AA, and Kappa of 75.70%, 84.66%, and 72.43%, respectively. The IP dataset has an unbalanced distribution of sample sizes per class and also contains diverse similar spectral vegetation, such as Corn class (classes 2–4), Grass class (classes 5–7), and Soybean class (classes 10–12). Therefore, distinguishing similar classes using the RBF-SVM method relying only on spectral features is difficult. The DL-based 3DCNN and SSRN methods improve the classification performance to a certain extent by exploring the deep spatial–spectral features. DFSL first introduced the FSL approach to tackle the HSI classification problems under sparse labeled samples, and the OA, AA, and Kappa are improved by 6.93%, 9.12%, and 8.06%, respectively, compared with the traditional 3DCNN. RN-FSC performs slightly worse than DFSL, especially in the Soybean class (classes 10 and 12). The DCFSL, Gia-CFSL, and CMFSL introduced domain adaptation tactics into their respective work to further improve the classification accuracy. However, these tactics cannot avoid misclassification caused by the "different bodies with the same spectrum." The DMCM method utilizes a dual-adjustment mode to train the classification model and obtains the highest or second highest identification accuracy on 13 classes of the IP dataset. Compared with CMFSL, the DMCM improves OA, AA, and Kappa by 8.22%, 4.87%, and 9.03%, respectively, and the classification results are remarkably stable. Fig. 8 shows the classification maps using different methods on the IP dataset, and the region (marked by red circle) demonstrates that DMCM has fewer misclassified pixels in the same class.

2) Classification Maps and Results of the UP Dataset: The experimental results of the different classification methods on the UP dataset are presented in Table VII. The proposed DMCM has higher OA, AA, and Kappa of 86.76%, 84.52%, and 82.23%, respectively, compared with the comparison methods. Overall, the FSL-based methods outperformed traditional classification methods, highlighting the advantages of FSL in solving few-shot classification problems. Among all the compared methods, RBF-SVM and 3DCNN produce poor classification results due to the sparse samples, and their OA is below 66%. With the existence of the residual structure, SSRN increases OA, AA, and Kappa to 76.96%, 81.82%, and 71.18%, respectively. Fortunately, the FSL approach mitigates poor classification performance of traditional DL models with sparsely labeled samples by episodes. DFSL improves OA

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TABLE VII
CLASSIFICATION RESULTS OF VARIOUS MODELS ON THE UP DATASET

Label	RBF-SVM [14]	3DCNN [26]	SSRN [30]	DFSL [60]	RN-FSC [61]	DCFSL [63]	Gia-CFSL [64]	CMFSL [68]	DMCM
1	64.34	59.82	**95.44**	73.43	68.55	74.68	77.54	78.67	82.48
2	62.34	63.05	95.69	89.25	93.44	84.79	86.90	86.33	**96.67**
3	58.60	**68.91**	58.36	48.09	49.81	60.64	60.16	67.05	55.24
4	93.06	77.31	**96.38**	84.72	92.15	89.02	94.24	93.34	86.05
5	93.13	90.77	**99.97**	99.65	99.43	99.46	98.78	99.61	98.38
6	44.23	63.40	40.84	67.81	57.99	**75.89**	70.52	74.35	65.01
7	90.45	87.64	80.33	64.48	70.04	78.75	80.63	78.23	**94.98**
8	62.74	57.27	72.47	67.37	63.48	67.82	70.74	79.41	**81.86**
9	99.77	95.57	98.74	92.92	99.19	99.21	98.76	98.41	**99.96**
OA	65.23±3.00	65.74±1.77	76.96±5.23	79.63±1.09	80.19±2.02	80.42±3.64	81.79±1.89	83.13±2.62	**86.76±0.80**
AA	74.29±2.05	73.72±1.01	81.82±3.37	76.41±1.39	77.12±1.33	81.14±1.11	82.23±1.56	83.93±1.71	**84.52±0.95**
Kappa	56.36±3.57	57.37±1.97	71.18±6.72	73.05±1.60	73.73±1.53	74.71±4.21	76.29±2.31	78.11±3.11	**82.23±1.10**

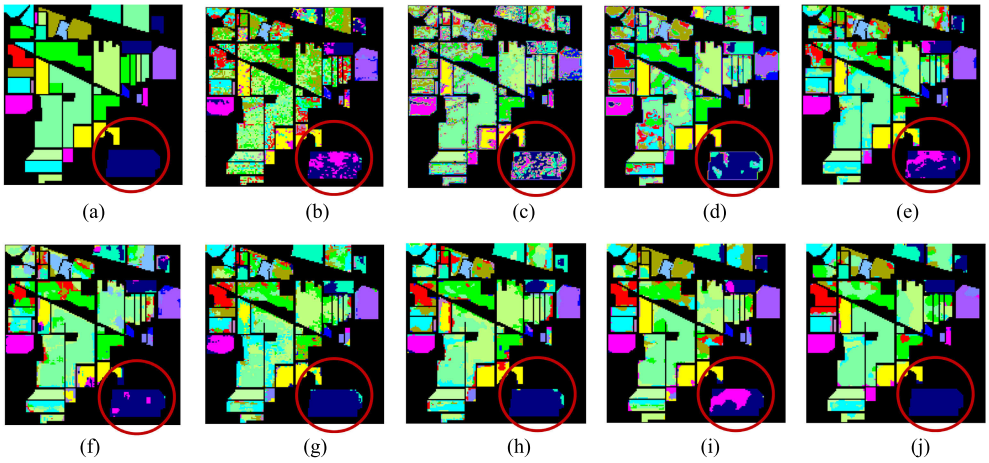


Fig. 8. Classification visualization maps of different models on the IP dataset. (a) Ground-truth map. (b) RBF-SVM. (c) 3DCNN. (d) SSRN. (e) DFSL. (f) RN-FSC. (g) DCFSL. (h) Gia-CFSL. (i) CMFSL. (j) DMCM.

by 2.67% compared with SSRN through the learned metric space. RN-FSC adopts a relational network and task-based learning strategy to break the OA value of the UP dataset by 80%. Nevertheless, DFSL and RN-FSC ignore the differences in data distribution across domains, thus leaving a room for advancement. For the DCFSL methods, CMFSL improved the OA value by 2.71% and 1.34% compared with the excellent DCFSL and Gia-CFSL, respectively. The DMCM method has the highest or second highest accuracy on five categories of the UP dataset; the OA, AA, and Kappa increased by 3.63%, 0.59%, and 4.12% compared with the superior CMFSL, respectively, and DMCM has small standard deviation. Fig. 9 also shows a smooth classification map, wherein fewer noise points are achieved with DMCM.

3) Classification Maps and Results of the PC Dataset: The classification results of the various methods on the PC dataset are shown in Table VIII. Compared with the current state-of-the-art (SOTA) methods, the DMCM framework has excellent and robust classification performance, achieving 98.00%, 95.11%, and 97.18% for OA, AA, and Kappa, respectively. The PC dataset has a relatively concentrated distribution of the same class; thus, the RBF-SVM, 3DCNN, and SSRN can achieve desirable classification results. In particular, SSRN shows strong classification competitiveness on the PC dataset, reaching 96.62%, 90.86%, and 95.24% for OA, AA, and Kappa, respectively. Despite slightly better OA and Kappa values of SSRN than DFSL and RN-FSC, poor classification results are observed for individual categories [e.g., self-blocking bricks (class 4) and tiles (class 6)]. The DMCM method obtains massive discriminative spatial spectral features through TGAN and achieves the highest or second highest classification accuracy on five classes on the PC dataset. Compared with the excellent SSRN, DMCM increases OA, AA, and Kappa by 1.38%, 4.25%, and 1.94%, respectively. Furthermore, the OA is improved by 0.59%, 1.06%, and 0.59% compared with the current popular DCFSL methods, DCFSL, Gia-CFSL, and CMFSL, respectively. Notably, Fig. 10 displays that the DMCM classification map generally has fewer misclassification points, with most areas intact and close to the original ground-truth map.

IV. DISCUSSIONS

A. Investigation of the Parameter Sensitivity

This section focuses first on the sensitivity analysis of two variables in TGAN: the kernel numbers N of the 3-D convolution and the scaling factors r in the ghost module. Fig. 11 illustrates the target dataset classification results when N is chosen in $\{4, 8, 16, 32\}$. Overall, as the number of convolutional kernels N increases, the classification results mainly tend to increase first and then decrease. The DMCM

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TABLE VIII
CLASSIFICATION RESULTS OF VARIOUS MODELS ON THE PC DATASET

Label	RBF-SVM [14]	3DCNN [26]	SSRN [30]	DFSL [60]	RN-FSC [61]	DCFSL [63]	Gia-CFSL [64]	CMFSL [68]	DMCM
1	99.34	98.31	**99.98**	99.10	99.64	99.79	99.62	99.62	99.79
2	76.27	82.29	94.81	89.93	94.48	**97.74**	90.63	91.64	93.95
3	79.82	86.63	85.64	93.03	94.10	93.84	94.10	92.77	**94.86**
4	75.87	78.86	68.50	83.73	**93.58**	92.15	93.09	92.75	89.29
5	79.91	87.01	96.04	80.83	84.25	90.16	88.14	91.02	**97.48**
6	93.35	83.60	82.61	90.83	89.83	96.72	94.73	96.04	**97.76**
7	77.41	82.16	**91.48**	83.25	81.44	84.05	84.42	86.80	84.71
8	86.26	96.12	**99.91**	97.71	97.43	98.81	98.24	98.64	99.07
9	**99.97**	99.56	98.73	95.73	98.53	98.31	98.33	99.04	99.07
OA	91.24±2.83	94.07±1.15	96.62±0.70	95.65±1.04	96.30±0.68	97.41±0.44	96.94±0.58	97.41±0.25	**98.00±0.17**
AA	85.36±3.36	88.28±3.63	90.86±1.29	90.46±2.52	92.59±1.11	93.95±1.00	93.48±0.84	94.26±0.65	**95.11±0.24**
Kappa	87.82±3.78	91.66±1.60	95.24±0.98	93.87±1.46	94.78±0.95	96.34±0.62	95.68±0.82	96.35±0.35	**97.18±0.25**

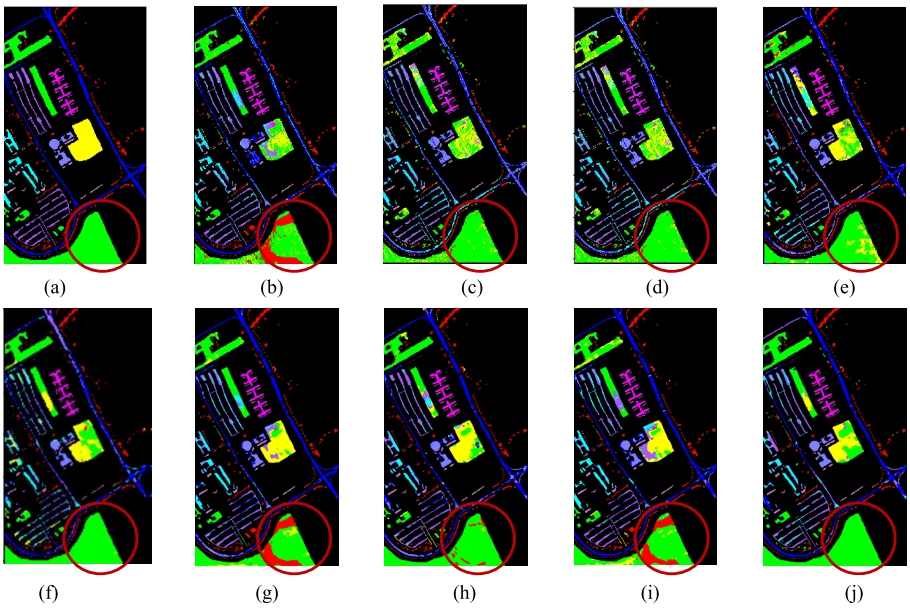


Fig. 9. Classification visualization maps of different models on the UP dataset. (a) Ground-truth map. (b) RBF-SVM. (c) 3DCNN. (d) SSRN. (e) DFSL. (f) RN-FSC. (g) DCFSL. (h) Gia-CFSL. (i) CMFSL. (j) DMCM.

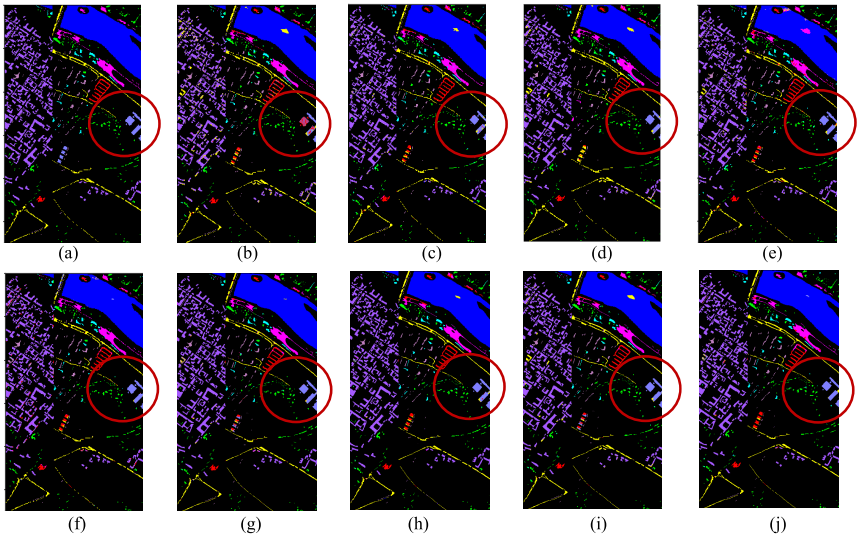


Fig. 10. Classification visualization maps of different models on the PC dataset. (a) Ground-truth map. (b) RBF-SVM. (c) 3DCNN. (d) SSRN. (e) DFSL. (f) RN-FSC. (g) DCFSL. (h) Gia-CFSL. (i) CMFSL. (j) DMCM.

performs best on the IP, UP, and PC datasets when $N = 8$. Large N values indicate that the model can gain additional features. However, the increase in the number of parameters and calculation cost will also affect the model capacity. Consequently, this article sets the N value to 8 consistently for usefulness and efficiency. After investigating the number

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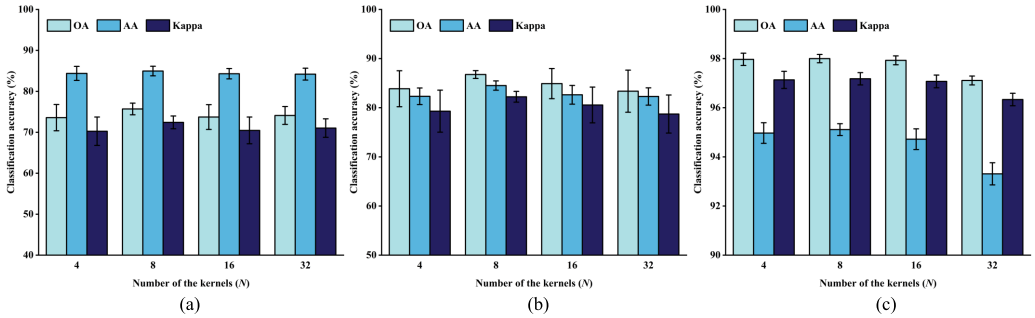


Fig. 11. Quantitative evaluation of different convolution kernel numbers N. (a) IP dataset. (b) UP dataset. (c) PC dataset.

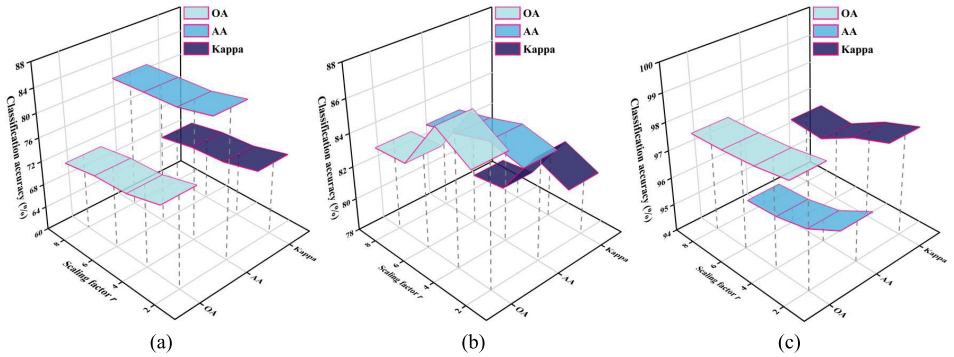


Fig. 12. Quantitative evaluation of different scaling factors r. (a) IP dataset. (b) UP dataset. (c) PC dataset.

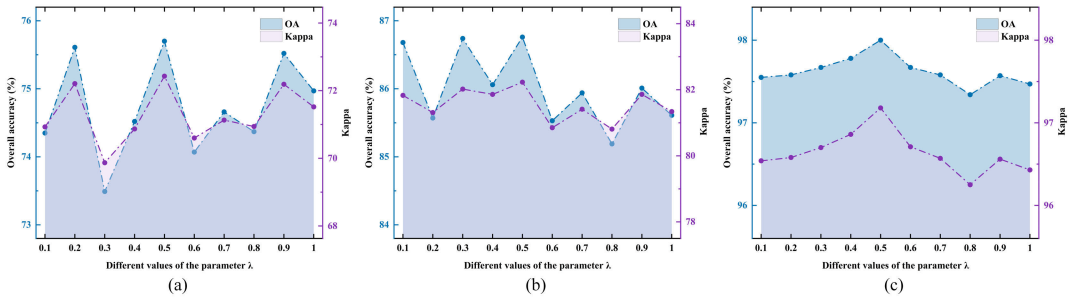


Fig. 13. Quantitative evaluation of parameter λ for the proposed DMCM. (a) IP dataset. (b) UP dataset. (c) PC dataset.

of convolutional kernels in TGAN, $N = 8$ is maintained, and the other hyperparameter r is adjusted in the range $\{2, 4, 6, 8\}$. A larger r drives substantial network compression and less computational loss. The classification accuracy band plots for different scaling factors r on different target datasets are displayed in Fig. 12. The classification accuracy gradually decreases when r is increased on the IP and PC datasets. In particular, DMCM obtains superior performance when $r = 2$. Moreover, an increase followed by a decrease is observed, as r increases on the UP dataset, and a classification accuracy watershed occurs at $r = 4$. Therefore, r is set to 2 on the IP and PC datasets, and that on the UP dataset is set to 4.

In addition, Fig. 13 exhibits the classification accuracy results of the three target datasets when λ is selected in $\{0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9, 1\}$. As can be seen from Fig. 13(a) and (c), the optimal value of the parameter λ is 0.5 on both IP and SA datasets. For the UP dataset, the optimal value of λ is 0.5 through taking the performance of DMCM on OA and Kappa into account.

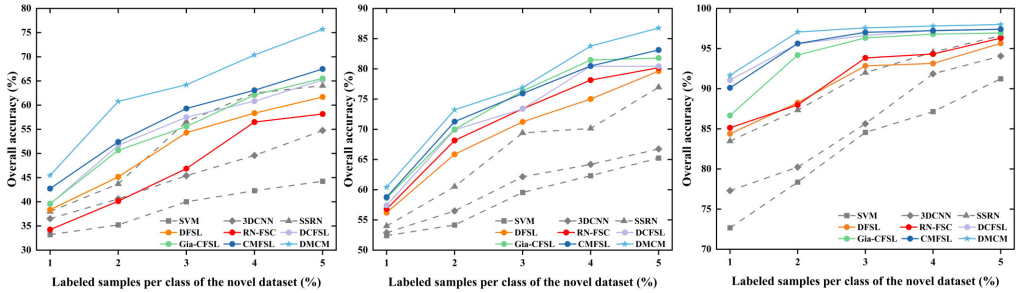
B. Ablation Study

Ablation experiments were conducted for each core module to verify the effectiveness of the DMCM method proposed in this article: different metrics, GA module, and IC and IA learning strategies. The baseline model adopted in this section comprises a regular 3-D convolutional network without any attention mechanism and follows the paradigm of meta-learning training. As shown in Table IX, when the GA module is introduced to obtain additional discriminative features using cheap operations, OA improves by 2.19%, 2.03%, and 1.05% on IP, UP, and PC, respectively. In addition, the traditional Euclidean distance metric is insensitive to intraclass samples. Thus, a CCM is introduced to consider the distribution of the feature space per class, improving the OA on IP, UP, and PC to 70.66%, 83.65%, and 97.47%, respectively. Moreover,

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TABLE IX
EVALUATION ON THE CONTRIBUTION OF VARIOUS MODULES ON DIFFERENT DATASETS

Euclidean	Class-Covariance	GA module	SC	IA	IP		UP		PC	
					OA	Kappa	OA	Kappa	OA	Kappa
✓	×	×	×	×	66.65±2.48	62.41±2.56	80.19±3.73	75.73±4.14	95.93±0.15	95.07±0.21
✓	×	✓	×	×	68.84±1.52	64.56±1.61	82.22±1.79	77.03±2.10	96.98±0.57	95.74±0.81
×	✓	✓	×	×	70.66±2.08	66.71±2.31	83.65±4.18	79.01±4.82	97.47±0.26	96.43±0.37
×	✓	✓	✓	×	73.35±1.95	69.70±2.09	84.42±4.29	80.00±4.92	97.87±0.36	96.65±0.50
×	✓	✓	✓	✓	**75.70±1.42**	**72.43±1.56**	**86.76±0.80**	**82.23±1.10**	**98.00±0.17**	**97.18±0.25**



(a) (b) (c)

Fig. 14. Comparison of OA under different target labeled sample numbers. (a) IP dataset. (b) UP dataset. (c) PC dataset.

the increased accuracy following the successive additions of IC and IA proves their effectiveness. Overall, the modules and learning mechanisms mentioned above enable effective improvement of the generalization and classification performance of the model under limited labeled samples.

C. Analysis of the Number of Target Labeled Samples

The amount of supervised information supplied during the training process is one of the critical factors affecting the classification accuracy. In this section, L labeled data are randomly selected from each class of target-domain datasets to observe the performance of various methods on OA, where L takes values from the range $\{1, 2, 3, 4, 5\}$. The results of the OA variation in IP, UP, and PC using different sample numbers are presented in Fig. 14. As expected, the OA increases for all classification models, as the L value grows. Constrained by the characteristics of traditional CNN learning, a perfect classification model usually demands the support of adequate labeled samples. Fig. 14 shows that SVM, 3DCNN, and SSRN are generally worse than other meta-learning-based methods, especially in the case of sparse samples. Furthermore, DFSL and RN-FSC ignore the domain shift problem, resulting in weaker classification performance than DCFSL, Gia-CFSL, and CMFSL. Significantly, the proposed DMCM method has better classification strengths than other comparison methods in any case.

D. Impact of Different Source-Domain Datasets on Classification Performance

Comparative experiments were conducted on the Dioni, Loukia, and Botswana datasets to further explore the impact of selecting different source datasets on few-shot HSI classification results. The HyRANK dataset (Dioni and Loukia datasets) was developed with the support of the International Society for Photogrammetry and Remote Sensing (ISPRS) Scientific Initiatives. Dioni and Loukia were collected by the Hyperion sensor (EO-1, USGS) with the spatial sizes of 250 × 1376 pixels and 249 × 945 pixels, respectively, containing 176 bands and the same 12 classes. The Botswana dataset captured by the EO-1 satellite of NASA has a spatial size of 1476 × 256 pixels and 14 classes, revealing 145 bands after removing the noise bands for training. Specifically, the number of labeled target samples for each class is uniformly set to five. Table X presents the classification results on novel datasets using different source-domain datasets. The overall classification performance of the DMCM is superior when the Chikusei is applied as source dataset. This performance may be attributed to the additional classes of Chikusei, covering various vegetation, soil, and man-made objects. Therefore, the diversity and richness of the source-domain samples are crucial to the model performance, especially in the field of few-shot HSI classification.

TABLE X
CLASSIFICATION RESULTS OF VARIOUS SOURCE-DOMAIN DATASETS ON NOVEL DATASETS

Target datasets	Metrics	Different source datasets			
		Dioni	Loukia	Botswana	Chikusei
IP	OA	74.05±1.18	74.53±1.41	74.82±0.94	**75.70±1.42**
	AA	83.53±1.05	84.36±0.99	83.91±0.89	**84.66±1.17**
	Kappa	70.59±1.35	71.15±1.56	71.42±1.06	**72.43±1.56**
UP	OA	85.52±0.87	**87.15±1.01**	86.05±0.78	86.76±0.80
	AA	83.24±1.13	83.99±0.96	83.66±0.93	**84.52±0.95**
	Kappa	80.59±1.20	**82.70±1.39**	81.22±1.06	82.23±1.10
PC	OA	97.71±0.24	97.87±0.18	97.73±0.20	**98.00±0.17**
	AA	94.46±0.48	94.68±0.46	94.72±0.38	**95.11±0.24**
	Kappa	96.76±0.34	96.99±0.25	96.79±0.29	**97.18±0.25**

E. Computational Complexity Evaluation

The numbers of floating-point operations (FLOPs), trainable parameters, and computational time are applied to demonstrate

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TABLE XI
FLOPS, PARAMETERS, AND COMPUTATIONAL TIME FOR THE DIFFERENT METHODS ON THREE DATASETS (FIVE LABELED SAMPLES PER CLASS GIVEN)

Datasets	Metrics	RBF-SVM	3DCNN	SSRN	DFSL	RN-FSC	DCFSL	Gia-CFSL	CMFSL	DMCM
IP	FLOPs (M)	-	0.40	1.73	41.14	6.25	46.98	41.29	29.27	7.33
	Params (M)	-	0.10	0.35	0.03	0.19	4.27	0.31	0.78	0.20
	Training time (s)	0.25	62.35	145.28	475.33	450.34	1835.17	8090.78	1319.41	3322.68
	Testing time (s)	0.43	1.5	7.6	4.77	42.69	1.14	1.33	0.85	0.93
UP	FLOPs (M)	-	0.20	0.99	41.14	6.25	46.19	41.29	29.27	7.98
	Params (M)	-	0.05	0.20	0.03	0.19	4.26	0.31	0.78	0.20
	Training time (s)	0.31	53.86	66.56	650.86	310.28	2207.59	5165.04	698.63	2830.82
	Testing time (s)	0.65	6.23	30.32	5.02	46.29	6.05	10.93	2.63	3.86
PC	FLOPs (M)	-	0.20	0.98	41.14	6.25	46.18	41.29	29.27	7.33
	Params (M)	-	0.05	0.20	0.03	0.19	4.26	0.31	0.78	0.20
	Training time (s)	0.29	54.39	66.49	687.72	391.50	2432.10	4643.44	792.94	1900.91
	Testing time (s)	0.65	21.50	106.37	16.59	72.97	23.47	28.22	9.57	10.39

the model complexity of the different methods. Meanwhile, the experiments are conducted with five labeled samples, and the details are shown in Table XI. Table XI reveals that RBF-SVM, 3DCNN, and SSRN possess less complexity compared with the FSL-based methods but are still unsuitable for identifying new classes under scarce labeled sample numbers. For the FSL-based methods, DCFSL suffers from high FLOPs and parameters due to the existence of the deep residual structure. Compared with the current popular DCFSL and Gia-CFSL algorithms, CMFSL performs well in both classification accuracy and computation time. Notably, the proposed DMCM framework adopts cheap operations to obtain additional features without massive FLOPs and parameters. Although DMCM costs relatively more training time compared to CMFSL, DMCM achieves higher and more stable classification results with similar testing time.

V. CONCLUSION

DMCM method is proposed in this study for few-shot HSI classification, which combines a low-parametric TGAN embedding feature extraction structure with an effective dual-adjustment mode comprising IC and IA learning strategies. First, the GA modules comprise ghost cheap operations, and NAMs are employed to construct the feature extraction framework TGAN, which obtains highly discriminative embedding features through low-computational waste. Then, to address the significant performance degradation experienced in cross-domain HSI classification, a dual-adjustment mode is adopted. In particular, this mode involves using an IC approach to regulate the accurate generation of class prototypes and an IA strategy to mitigate the difference in data distribution between the source and target domains. In addition, the CCM is integrated into few-shot HSI classification to optimize the decision boundary and further enhance the classification performance by considering the feature space distribution of each class. Extensive experiments and analyses on three widely used benchmark HSI datasets indicate that the proposed DMCM has strong generalization performance and superior classification capability compared with current SOTA methods, even in the presence of sparse labeled samples. Highly effective domain adaptation methods will be investigated in future work to overcome the limitations of fewer labeled samples, such as one-shot and zero-shot learning challenges.

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